**RESEARCH AND EDUCATIONAL USE ONLY — NOT A MEDICAL DEVICE**

# Pharmacogenomic analysis summary

| | |
|---|---|
| sample | DEMO_CONFIDENT |
| analysed | 2026-11-06T01:12:32.635356Z |
| printed | 2026-11-06 |
| drugs checked | 1 |

**This is not a clinical report.** It was produced by a final-year student project for research and education. It has not been clinically validated, and no qualified clinical expert has reviewed the explanatory text on this page.

Recommendation text on this page is quoted from published CPIC guidance via PharmCAT. Automated source checks are not human review: they cannot show that every statement here is supported, or that any of it is correct for the person this sample came from.

Do not change a medicine on the strength of this page. Take it to a qualified healthcare professional.

## Results

Ordered by consequence, not alphabetically.

### clopidogrel **Ineffective**

| | |
|---|---|
| gene | CYP2C19 |
| diplotype | *2/*2 |
| phenotype | PM |
| evidence level | Unknown |

clopidogrel: CYP2C19 *2/*2 (Poor Metabolizer) — Ineffective.

CPIC GUIDELINE — QUOTED EXACTLY

Avoid clopidogrel if possible. Use prasugrel or ticagrelor at standard dose if no contraindication.

CPIC strength of recommendation: Strong. Population: CVI ACS PCI. Implications: CYP2C19: Significantly reduced clopidogrel active metabolite formation; increased on-treatment platelet reactivity; increased risk for adverse cardiac and cerebrovascular events

— Annotation of CPIC Guideline for clopidogrel and CYP2C19 (via PharmCAT) [label rule: avoid_for_lack_of_efficacy]

## Unknown results

None — every drug checked returned a result.

## Input coverage

How many of the positions each gene needs carried a usable genotype in the uploaded file. A genotype caller reads an absent position as matching the reference, so missing data can produce confident results rather than uncertain ones — which is why the count is printed whether or not it passed, and why a gene below its minimum is reported as Unknown.

| gene | usable | required | minimum | sufficient |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| CYP2C19 | 35 | 35 | 100% | yes |
| CYP2C9 | 88 | 88 | 100% | yes |
| CYP2D6 | 157 | 157 | 100% | yes |
| CYP4F2 | 0 | 20 | 100% | NO |
| DPYD | 83 | 83 | 20% | yes |
| NUDT15 | 20 | 20 | 80% | yes |
| SLCO1B1 | 35 | 35 | 100% | yes |
| TPMT | 45 | 45 | 80% | yes |
| VKORC1 | 0 | 1 | 100% | NO |

**Not resolved from this file:** CYP4F2, VKORC1. Any drug keyed to these genes was reported as Unknown rather than assumed normal.

## Where this guidance came from

| | |
|---|---|
| PharmCAT | 3.4.0 |
| CPIC data version | 2026-07-13-11-40 |
| CPIC table SHA-256 | 5010bb813ed6aaacf504c689e1f74924ddc7c58383d5e35604074d299345d6a8 |
| allele definitions | 2026-07-13-11-40 |
| definitions SHA-256 | 7044c87682de18646a74b890739e2cc145ef0a3a21718a17b371dc013c7afe8f |
| explanations generated | 24 Jul 2026 |
| source | CPIC guidelines, retrieved via PharmCAT |

Guidance reflects what CPIC published when this data was captured. CPIC revises its guidelines. A weekly repository check compares this build's CPIC data with CPIC's current release and flags differences for review; an analysis does not check at the time it runs.